

Student: _____ Student ID#: _____

EXAM 2

1. The answers to the questions in this exam must be your own. **Violations will result in a Fail grade in the course** and will be handled according to NAU's Academic Integrity Policy.
2. Sign this first page with your name and NAU id **before the test starts**.
3. Read the instructions and questions carefully. Correctly interpreting the questions is part of the examination. **I will NOT answer questions during the exam.**
4. The exam starts at 10:00 AM and ends at 12:00 PM. If you write in your exam after time runs up, your exam may be taken from you, and you'll receive a ZERO as a grade.
5. Phones, laptops, or any other electronic devices are not allowed during exam time. Please turn your phones off and store them out of your reach. If you use any devices during exam time, the exam may be taken from you, and you'll receive a zero as a grade.
6. Provide your answer using a non-erasable pen. If you use pencil or erasable pen, you cannot request grade review.
7. The exam is **individual** and closed book, closed notes.
8. Your grade will appear on BBLearn as soon as all the tests are graded. Don't email me asking for grades, if it's not on BBLearn, I don't have them just yet. Thanks for your patience.
9. There is **only one correct answer** for each question.
10. **No partial marks** are possible in the questions.

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1. (4 points) About the basics of programming languages

- I. Programming languages are designed only to enable the communication among developers
- II. Programming languages are more expressive than natural language
- III. In the beginning the programming languages were closer to machine-level commands and later they evolved to enable developers' readability

(a) **Only III is correct**

(b) Only I and II are correct

(c) All the items are correct

(d) Only items I and III are correct

(e) II and III are correct

2. (4 points) About the evaluation criteria for choosing a programming language, mark the correct sentence:

- (a) Reliability of a programming language is closely related to the level of support that the community offers
- (b) To make a programming language simpler it is necessary to make it closer to natural language, increasing the number of basic constructs
- (c) Too much orthogonality is good to increase the writability
- (d) **Support is an important criterion to choose a programming language**
- (e) Since programming languages are usually open source, cost is not a problem when choosing a programming language

3. (8 points) Match the columns add the corresponding letter on the left to the right column

A. Interpreted Languages

B. Compiled Languages

A	Debug the source code at run time, line by line
B	The code generated is specific for a hardware architecture
B	A binary code is generated and the program is executed from this binary
B	The execution is, in general, faster
A	The program needs to be executed from the source code every time
B	C is an example of this type of languages
A	The code is generated dynamically and can be executed piece by piece
A	PHP and Python are examples of this type of language

4. (4 points) About the historical perspective of Programming Languages, check correct sentence

- (a) Machine language-based languages were the first ones, usually having commands that were easier for developers understand
- (b) The von Neumann architecture was not considered when programming languages were created
- (c) **High-level languages were introduced to make the maintenance easier**
- (d) COBOL was created to deal with the scientific problems at the 2000's

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(e) All the languages we use currently are from the 50's, 60's, or 70's, no new languages had been created after that.

5. (8 points) Consider the following grammar:

$\langle S \rangle \rightarrow \langle A \rangle a \langle B \rangle b$

$\langle A \rangle \rightarrow \langle A \rangle b \mid b$

$\langle B \rangle \rightarrow a \langle B \rangle \mid a$

Are these sentences part of the language defined by the grammar? Show the leftmost derivation for each of them (step-by-step, changing one symbol per derivation) and conclude if they are part of the language or not. **Answers with no derivation will receive ZERO.**

baab $\langle S \rangle$ $\langle A \rangle$ a $\langle B \rangle$b b a $\langle B \rangle$b b a a b <i>The sentence is part of the language</i>	bbbab $\langle S \rangle$ $\langle A \rangle$ a $\langle B \rangle$b $\langle A \rangle$b a $\langle B \rangle$b $\langle A \rangle$bb a $\langle B \rangle$b b bb a $\langle B \rangle$b b bb a ??? <i>Impossible to do anything other than add at least one more "a" derived from $\langle B \rangle$</i>
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6. (4 points) Consider the following sentences about carbon footprint/green IT and programming languages:

- I. The resources (memory, computation time) used by the programming language influence on the energy consumption and, therefore, the carbon footprint
- II. The carbon footprint is influenced by the tools the hardware used and programming languages do not have influence on that
- III. Python is very efficient, consuming way less energy than C, for example

we can say that:

- (a) Only item I is correct
- (b) Items I and II are correct, and III is incorrect
- (c) Items I and III are correct, II is not
- (d) All items are correct
- (e) Items II and III are correct, but item I is not

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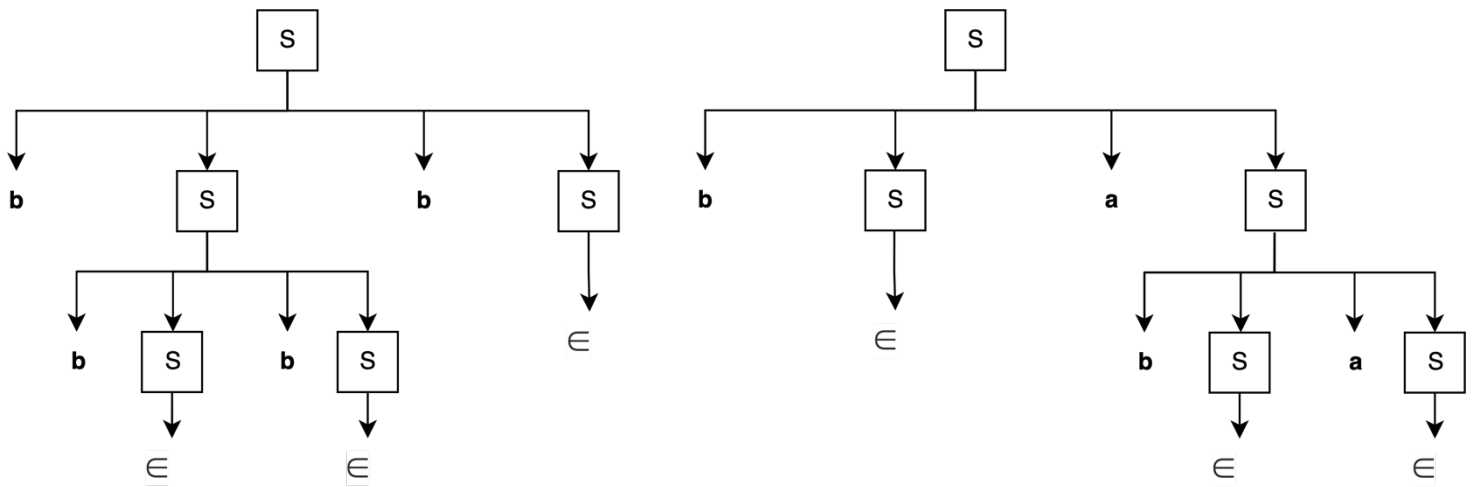
7. (8 points) Prove that the following grammar is ambiguous

$S \rightarrow bSaS$

$S \rightarrow SaSb$

$S \rightarrow \epsilon$

It is possible to derive 2 different trees for the word **baba**:



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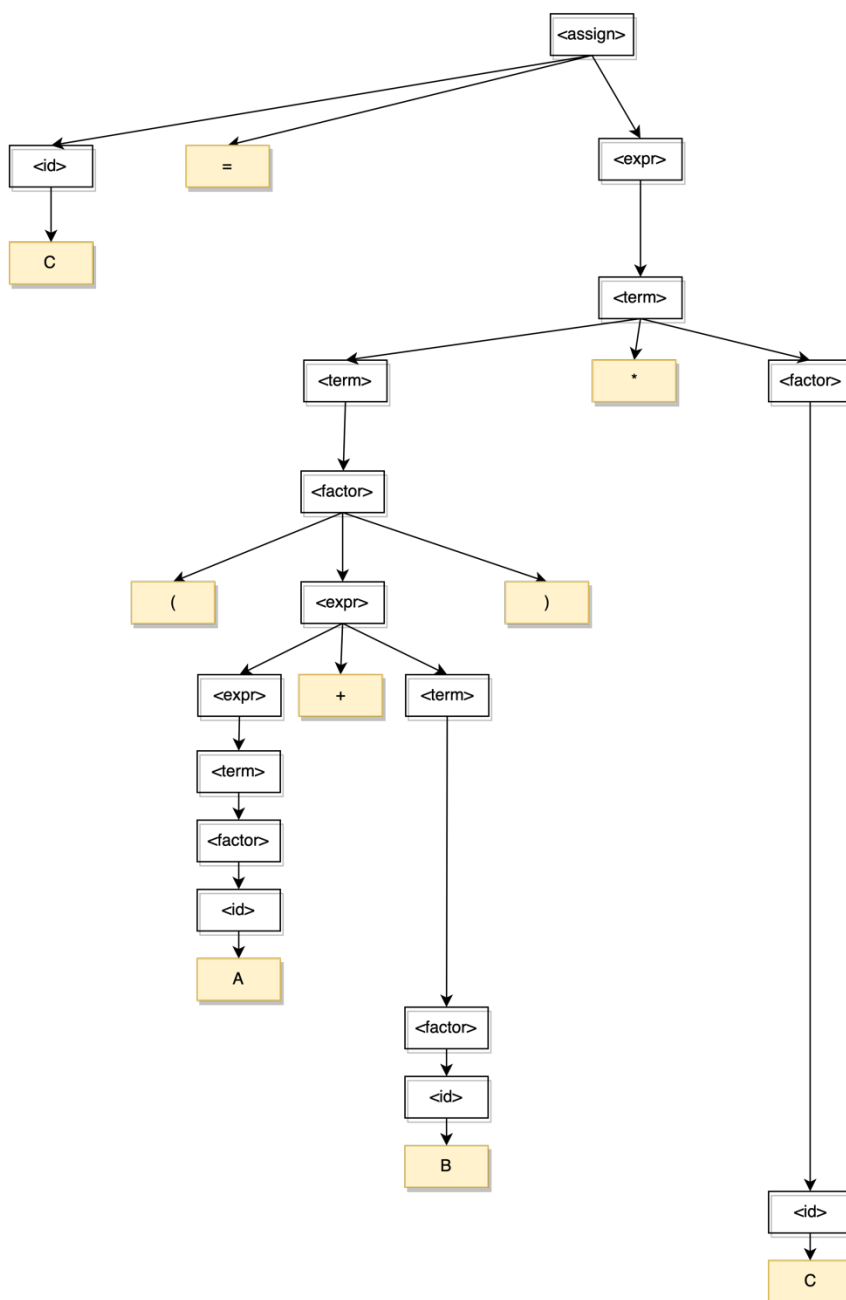
8. (8 points) Given the following grammar

```

<assign>  →  <id> = <expr>
<id>      →  A |
              B |
              C
<expr>    →  <expr> + <term> |
              <term>
<term>    →  <term> * <factor> |
              <factor>
<factor>  →  ( <expr> ) |
              <id>
    
```

Create the parse tree considering for the grammar and sentence below:

C = (A + B) * C



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9. (4 points) About imperative programming, it is **incorrect** to say that:

- (a) It follows a process of describing steps that simply change the state of the computer.
- (b) It mainly focuses on describing how the program executes step-by-step.
- (c) These programs are structured as successive assignments of values to variable names.
- (d) It is generally a process of developing software simply by composing functions.**
- (e) It uses statements that change a program's state.

10. (8 points) Match the paradigm on the left with the sentences on the right

A. Imperative Programming	A	Based on the Von Neumann architecture model
B. Functional Programming	A	Based on a paradigm where each statement is sequentially evaluated
C. Logic Programming	B	Repetition happens only by recursion not by loops sequencing commands
	C	Prolog is the representative of this paradigm
	B	Generally structured as successive nested function calls.
	A	Uses statements that change a program's state.
	B	Referential Transparency is one of its characteristics
	C	Relies on facts and relations between objects

11. (4 points) Consider the following sentences about inheritance

- I. A subclass inherits only the functions from a superclass, not the variables
- II. Even private functions from a superclass are inherited by the subclass
- III. Multiple inheritance is possible, and each language have their own way avoid the diamond problem
- IV. One way of implementing polymorphism is by overriding a function from a superclass.

we can say that:

- (a) Only I and II are correct
- (b) I, II and III are correct, but IV is incorrect
- (c) I, II and IV are correct, but III is not
- (d) III and IV are the only correct sentences**
- (e) All the sentences are correct

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12. (4 points) About object-oriented programming, check the incorrect alternative

- (a) In Object Oriented programs it is possible to determine the visibility of a variables and function, varying from public (visible to any other object) to private (visible only inside of the object).
- (b) The object-orientated paradigm is based on the concept of object, which encapsulates both data and the functions to manipulate its data
- (c) Class is a model for a set of objects
- (d) Object orientation is one type of declarative programming languages**
- (e) Inheritance allows a class to reuse and extend the variables and functions of another class

13. (8 points) Given the following code in Haskell:

```
func :: Integer -> Integer
func n
  | n == 0 = 1
  | n == 1 = 1
  | otherwise = func (n-1) + func (n-2)
```

a. What is the output when we call `func 4`

5 (*This is an adaptation of the Fibonacci function presented in class*)

b. What does `Integer -> Integer` in the first line represent?

The domain and the range

OR: type of the input and type of return

14. (4 points) Given the following facts and rule in Prolog:

parent(a,b) . parent(c,b) . parent(a,d) . parent(b,e) . parent(b,f) .	grandchild(X,Y) :- parent(Y,Z) , parent(Z,X) .
---	--

What would be the result of the query:
`grandchild(G,a)`

**e
f**

e and f are the grandchild of a (a is parent of b, b is parent of e and f)

15. (4 points) Given the following code in Prolog:

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```
func(1, 1) .  
func(N, Result) :-  
    NEXT is N-1,  
    func(NEXT, Return),  
    Result is N * Return.
```

What is the output when we call `func(3, R)`

6 (*This is a factorial function that I presented in class*)

16. (4 points) About type binding and variable declaration:

- I. One advantage of dynamic type binding is that you can change the type of the variable at any point of the program
- II. Explicit declaration means that a program statement is used for declaring/defining the type of a variable
- III. In explicit declaration a variable receives a type when it is first declared, and the type never changes
- IV. Python used dynamic type binding and implicit declaration

we can say that:

- (a) Only I and II are correct**
- (b) I, II and III are correct, but IV is incorrect
- (c) II, III and IV are correct, but I is not
- (d) I and III are the only correct sentences
- (e) Only II and III are correct

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17. (4 points) Considering that this Python program is executed using **dynamic scoping**, what would be the output?

```
a = 5
b = 4

def f_a():
    b = 7
    print (a + b)
    c = f_b()
    return c

def f_b():
    b = 2
    return f_c()

def f_c():
    return a + b

print (f_a())
```

12

7

18. (8 points) Mark True (T) or False (F) for the following sentences about lexical and syntax analysis

- [T] The output of Lexical Analysis step a set of lexemes and entries in the Symbol Table
- [F] The lexical analyzer breaks sentences into lexemes (words) and identify the meaning of a complete sentence
- [F] To identify a token, the lexical analyzer will the complete line of a program
- [T] Comments are not defined as part of the grammar and are discarded before the lexical analyzer
- [T] After the lexical analyzer finishes its job, the syntax analysis takes place in which the tokens are analyzed giving an abstract syntax tree as output
- [T] The syntax analyzer checks the sequence of tokens to validate if the sequence in which they appear is correct
- [T] A syntax parse tree is built according to the grammar during the syntax analysis, considering the lexical units
- [F] An Abstract Syntax Tree (AST) is bigger than the Parse Tree because it adds "syntactic sugar" to the tree